Section A — Objective Type Questions

1. Which of the following combines with a male gamete to form the endosperm?
(A) Secondary nucleus (B) Synergid cell
(C) Antipodals (D) Unfertilized egg

2. Entomophily is pollination carried out by:
(A) Wind (B) Insects (C) Water (D) Birds

3. Anemophily is pollination carried out by:
(A) Birds (B) Bats (C) Wind (D) Snails

4. Which of the following is a dioecious plant?
(A) Mustard (B) Hibiscus (C) Pea (D) Papaya

5. What is the ploidy level of endosperm in angiosperms?
(A) $3n$ (B) $2n$ (C) n (D) $4n$

6. Wind-pollinated flowers are generally:
(A) Nectar-rich
(B) Small and colourless
(C) Large and attractive
(D) Fragrant

7. Pollen grains fundamentally represent the:
(A) Sporophyte (B) Female gametophyte
(C) Male gametophyte (D) Zygote

8. The wall of a pollen grain is generally:
(A) Single-layered (B) Three-layered
(C) Multi-layered (D) Two-layered

9. In which fruit is the aril edible?
(A) Litchi (B) Apple (C) Mango (D) Coconut

10. The central cell of the embryo sac contains the:
(A) Primary nucleus (B) Secondary nucleus
(C) Synergid (D) Egg cell

11. In which of the following plants does water pollination occur?
(A) Rose (B) Maize (C) Hydrilla (D) Papaya

12. In apple, fruit development primarily involves which floral part?
(A) Petals (B) Calyx
(C) Ovary (D) Thalamus

13. In cleistogamous flowers, there is assured:
(A) Self-pollination (B) Cross-pollination
(C) Artificial pollination (D) Insect pollination

14. Seed formation without fertilization in Asteraceae and grasses is called:
(A) Polyembryony (B) Apomixis
(C) Parthenocarpy (D) Parthenogenesis

15. The gynoecium of a flower is composed of:
(A) Anther and filament
(B) Sepals and petals
(C) Stigma, style and ovary
(D) Thalamus

16. Flowers of *Vallisneria* are:
(A) Anemophilous (B) Entomophilous
(C) Zoophilous (D) Hydrophilous

17. An embryo sac originates from which of the following structures?
(A) Megaspore (B) Microspore
(C) Microsporangium (D) Pollen tube

18. Tapetal cells of an anther are generally:
(A) Haploid (B) Polyploid/Multinucleate
(C) Diploid (D) Enucleate

19. Banana is a natural example of:
(A) Apomixis (B) Polyembryony
(C) Parthenocarpy (D) Parthenogenesis

20. Synergid cells present in the embryo sac are:
(A) Triploid (B) Polyploid (C) Diploid (D) Haploid

21. Autogamy is invariably assured in:
(A) Xenogamy (B) Cleistogamy
(C) Geitonogamy (D) Chasmogamy

22. Which of the following is NOT a part of the microsporangium?
(A) Epidermis (B) Endothecium (C) Micropyle (D) Tapetum

23. In angiosperms, which cell divides to form two male gametes?
(A) Vegetative cell (B) Microspore mother cell
(C) Megaspore (D) Generative cell

24. Which of the following is NOT associated with the egg apparatus?
(A) Antipodal cells (B) Egg cell
(C) Synergids (D) Filiform apparatus

25. The single cotyledon present in a grass seed is known as:
(A) Endosperm (B) Scutellum (C) Embryonal axis (D) Seed coat

26. A form of asexual reproduction that mimics sexual reproduction is:
(A) Parthenocarpy (B) Polyembryony
(C) Apomixis (D) Vegetative propagation

27. In which of the following seeds is the endosperm NOT completely consumed by the developing embryo?
(A) Pea (B) Groundnut (C) Bean (D) Coconut

28. Transfer of pollen grains from the anther to the stigma of another flower of the same plant is:
(A) Geitonogamy (B) Autogamy
(C) Xenogamy (D) Cleistogamy

29. Reproduction that does not involve gametic fusion is called:
(A) Double fertilization (B) Apomixis (C) Syngamy (D) Triple fusion

30. In flowering plants, the male reproductive whorl is represented by the:
(A) Gynoecium (B) Calyx (C) Androecium (D) Corolla

31. A typical stamen consists of which two parts?
(A) Stigma and style (B) Ovary and ovule
(C) Thalamus and petal (D) Anther and filament

32. A typical angiosperm anther is:
(A) Bilobed and dithecous (B) Tetralobed
(C) Monothecous (D) Unilobed

33. What is the total number of microsporangia in a typical angiosperm anther?
(A) Two (B) Four (C) Eight (D) One

34. The innermost wall layer of a microsporangium is the:
(A) Epidermis (B) Endothecium (C) Tapetum (D) Middle layers

35. The primary function of the tapetum is:
(A) Dehiscence of anther (B) Protection
(C) Photosynthesis (D) Nourishment of developing pollen grains

36. The process of formation of microspores from a pollen mother cell (PMC) is termed:
(A) Microsporogenesis (B) Megasporogenesis
(C) Embryogenesis (D) Parthenogenesis

37. What is the ploidy level of cells in a microspore tetrad?
(A) Diploid (B) Haploid (C) Triploid (D) Polyploid

38. The hard outer wall layer of a pollen grain is called:
(A) Intine (B) Epidermis (C) Exine (D) Endothecium

39. The exine of a pollen grain is composed of which highly resistant organic material?
(A) Cellulose (B) Pectin (C) Lignin (D) Sporopollenin

40. Apertures on the exine where sporopollenin is absent are called:
(A) Germ pores (B) Stomata (C) Hilum (D) Chalaza

41. The thin inner wall layer (intine) of a pollen grain is primarily made of:
(A) Lignin and suberin (B) Sporopollenin
(C) Cellulose and pectin (D) Chitin

42. The larger cell in a mature pollen grain having abundant food reserve is the:
(A) Generative cell (B) Synergid
(C) Antipodal (D) Vegetative cell

43. The small spindle-shaped cell floating in the cytoplasm of the vegetative cell is the:
(A) Generative cell (B) Egg cell (C) Male gamete (D) Central cell

44. In over 60% of angiosperms, pollen grains are shed at the:
(A) 1-celled stage (B) 2-celled stage (C) 3-celled stage (D) 4-celled stage

45. Which invasive weed entered India with imported wheat and causes severe pollen allergy?
(A) Water hyacinth (B) *Lantana* (C) *Parthenium* (Carrot grass) (D) *Hydrilla*

46. Pollen grains can be stored for years in pollen banks using:
(A) Formalin (B) Liquid oxygen
(C) Absolute alcohol (D) Liquid nitrogen ($-196^\circ C$)

47. A multicarpellary, syncarpous pistil is observed in:
(A) *Papaver* (Opium) (B) *Michelia* (C) Pea (D) Mango

48. A multicarpellary, apocarpous gynoecium is found in:
(A) *Papaver* (B) *Michelia* (C) Hibiscus (D) Tomato

49. The basal swollen part of the pistil is termed the:
(A) Stigma (B) Style (C) Ovary (D) Thalamus

50. Arising from the placenta inside the ovary are the:
(A) Styles (B) Stigmas (C) Petals (D) Megasporangia (Ovules)

51. The stalk that attaches an ovule to the placenta is the:
(A) Funicle (B) Micropyle (C) Chalaza (D) Hilum

52. The junction where the body of the ovule fuses with the funicle is called the:
(A) Chalaza (B) Hilum (C) Nucellus (D) Micropyle

53. The protective envelopes encircling the ovule are called:
(A) Nucellus (B) Pericarp
(C) Integuments (D) Tapetum

54. Integuments encircle the ovule except at a small opening at the tip called the:
(A) Chalaza (B) Hilum (C) Funicle (D) Micropyle

55. The basal region of the ovule located opposite to the micropylar end is the:
(A) Chalaza (B) Hilum (C) Nucellus (D) Funicle

56. The central mass of cells enclosed within the integuments with reserve food is the:
(A) Endosperm (B) Nucellus
(C) Embryo sac (D) Synergid

57. The female gametophyte of an angiosperm is also known as the:
(A) Endosperm (B) Nucellus (C) Embryo sac (D) Ovule

58. The division that a megaspore mother cell (MMC) undergoes to produce megaspores is:
(A) Mitosis (B) Amitosis (C) Multiple fission (D) Meiosis

59. Development of an embryo sac from a single functional megaspore is termed:
(A) Monosporic development (B) Bisporic development
(C) Apomixis (D) Polysporic development

60. In a majority of flowering plants, out of 4 megaspores, how many remain functional?
(A) Four (B) One (C) Three (D) Two

61. A typical mature angiosperm embryo sac is:
(A) 8-celled, 8-nucleate (B) 7-celled, 7-nucleate
(C) 8-celled, 7-nucleate (D) 7-celled, 8-nucleate

62. The group of three cells situated at the micropylar end is collectively termed the:
(A) Egg apparatus (B) Polar nuclei
(C) Central cell (D) Antipodals

63. The special cellular thickenings at the micropylar tip of synergids are called the:
(A) Egg cell (B) Filiform apparatus
(C) Central cell (D) Antipodal cells

64. The primary function of the filiform apparatus is to:
(A) Nourish the embryo
(B) Protect the egg cell
(C) Guide the entry of the pollen tube into the synergid
(D) Produce endosperm

65. The three cells located at the chalazal end of the embryo sac are called:
(A) Synergids (B) Egg cells
(C) Polar nuclei (D) Antipodals

66. The large central cell of an embryo sac contains:
(A) Two polar nuclei (B) Three antipodals
(C) One egg cell (D) Two synergids

67. Pollination involving transfer of pollen grains to the stigma of a genetically different plant is:
(A) Autogamy (B) Xenogamy (C) Geitonogamy (D) Cleistogamy

68. The defining characteristic of cleistogamous flowers is that:
(A) They open only at night (B) They require cross-pollination
(C) They do not open at all (D) They produce excessive nectar

69. Pollination in grasses and maize is predominantly carried out by:
(A) Water (B) Insects (C) Birds (D) Wind

70. In marine sea-grasses like *Zostera*, pollination is achieved via:
(A) Water (B) Insects (C) Birds (D) Wind

71. In emergent aquatic plants such as water hyacinth and water lily, pollination occurs by:
(A) Water currents (B) Insects or wind
(C) Bats (D) Submerged currents

72. Flowers pollinated by flies and beetles secrete foul odours in order to:
(A) Repel predators (B) Deter herbivores
(C) Attract these pollinators (D) Secrete nectar

73. The flower of *Amorphophallus* provides floral reward to pollinators in the form of:
(A) Endosperm for eating (B) Fresh water droplets
(C) Shelter in hollow stems (D) Safe place to lay eggs

74. The relationship between the *Yucca* plant and its pollinator moth is an example of:
(A) Obligate mutualism (B) Competition
(C) Commensalism (D) Parasitism

75. A genetic mechanism that prevents self-pollen from fertilizing the ovules is:
(A) Cleistogamy (B) Self-incompatibility
(C) Apomixis (D) Parthenocarpy

76. In papaya, male and female flowers are borne on separate plants (dioecy); this condition prevents:
(A) Only autogamy
(B) Only xenogamy
(C) Both autogamy and geitonogamy
(D) Only geitonogamy

77. The removal of anthers from a bisexual flower bud before dehiscence is called:
(A) Bagging (B) Tagging (C) Pollination (D) Emasculation

78. Covering emasculated flowers with a bag of butter paper to prevent contamination is called:
(A) Bagging (B) Hybridization (C) Dusting (D) Emasculation

79. After entering the embryo sac, the pollen tube discharges both male gametes into the:
(A) Antipodal cells (B) Cytoplasm of a synergid
(C) Egg cell cytoplasm (D) Central cell

80. The fusion of one male gamete with the egg cell nucleus is termed:
(A) Triple fusion (B) Parthenogenesis (C) Syngamy (D) Polyembryony

81. The fusion of the second male gamete with two polar nuclei is called:
(A) Triple fusion (B) Parthenocarpy (C) Syngamy (D) Double fertilization

82. Occurrence of both syngamy and triple fusion in an embryo sac is known as:
(A) Self-pollination (B) Double fertilization (C) Apomixis (D) Polyembryony

83. The triploid nucleus resulting from triple fusion is the:
(A) Zygote (B) Generative nucleus
(C) Primary endosperm nucleus (PEN) (D) Vegetative nucleus

84. Development of endosperm precedes embryo development because it:
(A) Forms the seed coat
(B) Guides the pollen tube
(C) Makes the fruit sweet
(D) Provides assured nutrition to developing embryo

85. The coconut water from tender coconut represents:
(A) Free-nuclear endosperm (B) Cellular endosperm
(C) Perisperm (D) Nucellus

86. In which of the following plants is endosperm completely consumed before seed maturity?
(A) Coconut (B) Pea (C) Wheat (D) Castor

87. The early stages of embryo development are collectively termed:
(A) Seed germination (B) Apomixis (C) Embryogeny (D) Parthenogenesis

88. The portion of the embryonal axis above the level of cotyledons is the:
(A) Hypocotyl (B) Radicle (C) Coleoptile (D) Epicotyl

89. The cylindrical portion below the level of cotyledons terminating in the radicle is the:
(A) Hypocotyl (B) Plumule (C) Coleorhiza (D) Epicotyl

90. In a monocot embryo, the hollow foliar structure enclosing the shoot apex is the:
(A) Coleorhiza (B) Coleoptile (C) Perisperm (D) Integument

91. In grass embryos, the undifferentiated sheath enclosing the root cap and radicle is the:
(A) Coleoptile (B) Scutellum (C) Coleorhiza (D) Seed coat

92. During seed maturation, the moisture content is reduced to roughly:
(A) 1--5% (B) 30--40% (C) 50--60% (D) 10--15% by mass

93. The residual, persistent nucellus present in some seeds is called:
(A) Perisperm (B) Endosperm (C) Pericarp (D) Seed coat

94. Following fertilization, the wall of the ovary transforms into the fruit wall called:
(A) Testa (B) Pericarp (C) Endosperm (D) Cotyledon

95. In fruits like apple, strawberry and cashew, fruit formation primarily involves the:
(A) Ovary wall (B) Petals (C) Thalamus (D) Calyx

96. Fruits that develop with contributions from floral parts other than the ovary are called:
(A) True fruits (B) Parthenocarpic fruits (C) Aggregate fruits (D) False fruits

97. The development of fruits without fertilization is referred to as:
(A) Parthenocarpy (B) Apomixis (C) Polyembryony (D) Parthenogenes

98. A viable seed of *Lupinus arcticus* excavated from Arctic Tundra germinated after dormancy of:
(A) 1,000 years (B) 10,000 years (C) 2,000 years (D) 50,000 years

99. The occurrence of more than one embryo in a single seed is known as:
(A) Apomixis (B) Parthenocarpy (C) Polyembryony (D) Amphimixis

100. Production of seeds without fertilization in flowering plants is termed:
(A) Polyembryony (B) Parthenocarpy (C) Amphimixis (D) Apomixis